
HYDRON
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AIDE

HYDRON

Improving performance and safety in the pool, with a better designed waterpolo cap to withstand tougher and rougher opponents; Improving protection around the ears while allowing for better hearing to enhance team and referee communication.

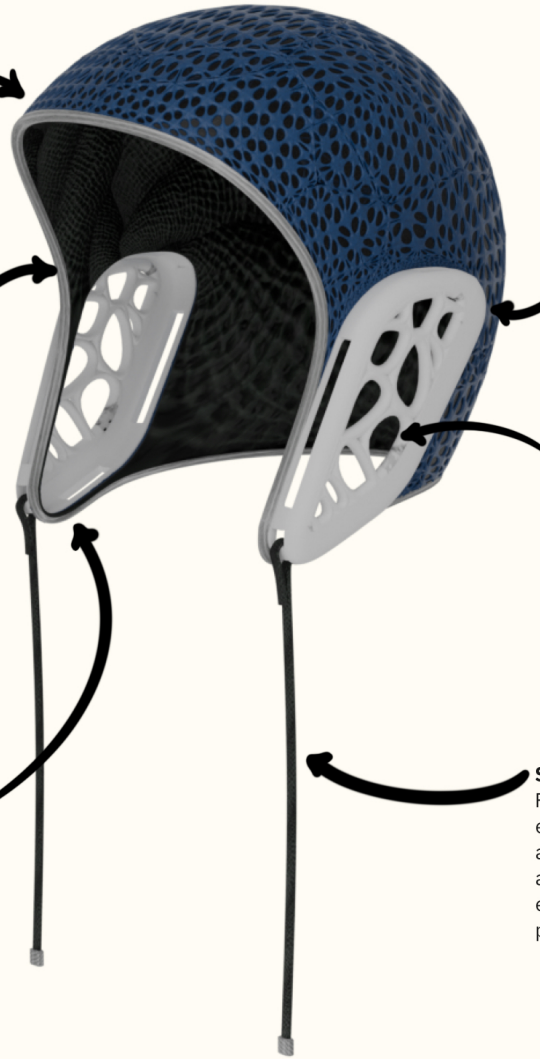


FEATURES

Silicone Mesh:
The silicone mesh acts two fold. Firstly, it improves protection for the user softening impacts and blows against the head, and secondly, it adds a flair to the design making it more interesting to look at then a simple white surface.

Soft Fabric:
A softer fabric on the inside of the cap helps to protect the user and makes the cap more comfortable to wear for long periods of time. The extra layer also stops the silicone from being directly on someone's skin which can be irritative.

Guard Padding:
Padding increases on the inside of the ear guards in the form of more dense silicone and the fabric, absorbing the impact from blows without having hard plastic directly on the skin.



Streamlined Guards:
Improving both intuitiveness and the hydrodynamics for the user, ensuring both greater comfort and performance. The shape gives even more room for the ears ensuring no snagging occours.

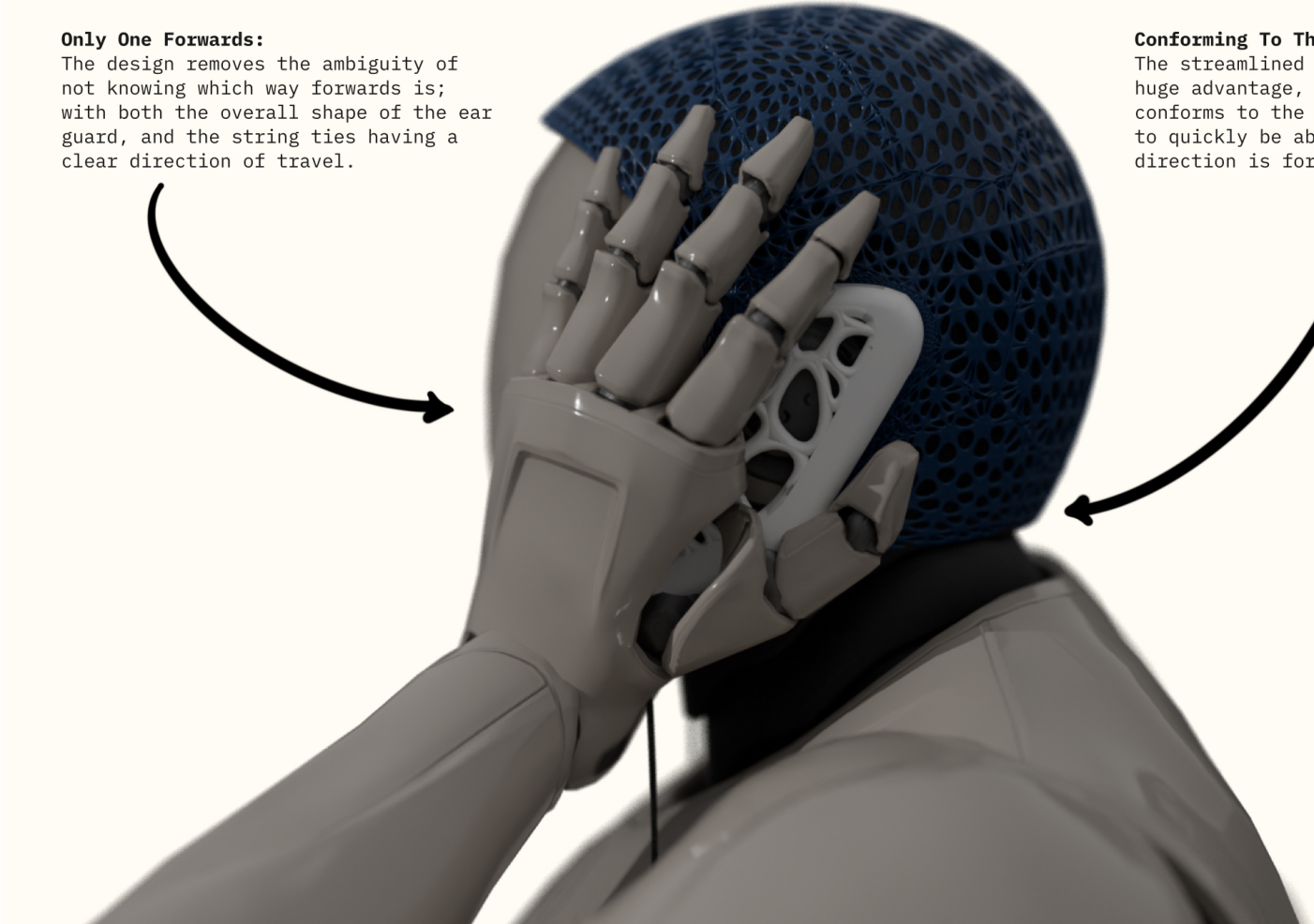
Improved Hearing:
Larger holes in a Voronoi mesh both improve looks and allow for more sound to pass through. Holes in the front and back of the ear guard allow for even more sound to pass through from both in front and behind a user.

String Ties:
Fitting with regulations, string ties ensure the most comfort and adaptability for the users. These are attached directly to the ear guards to ensure the cap is as secure as possible.

MADE TO BE INTUITIVE

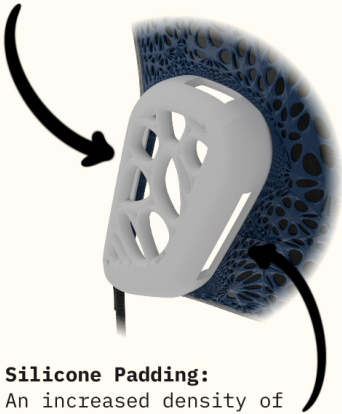
Only One Forwards:
The design removes the ambiguity of not knowing which way forwards is; with both the overall shape of the ear guard, and the string ties having a clear direction of travel.

Conforming To The Thumb:
The streamlined design has another huge advantage, where it easily conforms to the thumb allowing users to quickly be able to tell which direction is forwards for the cap.

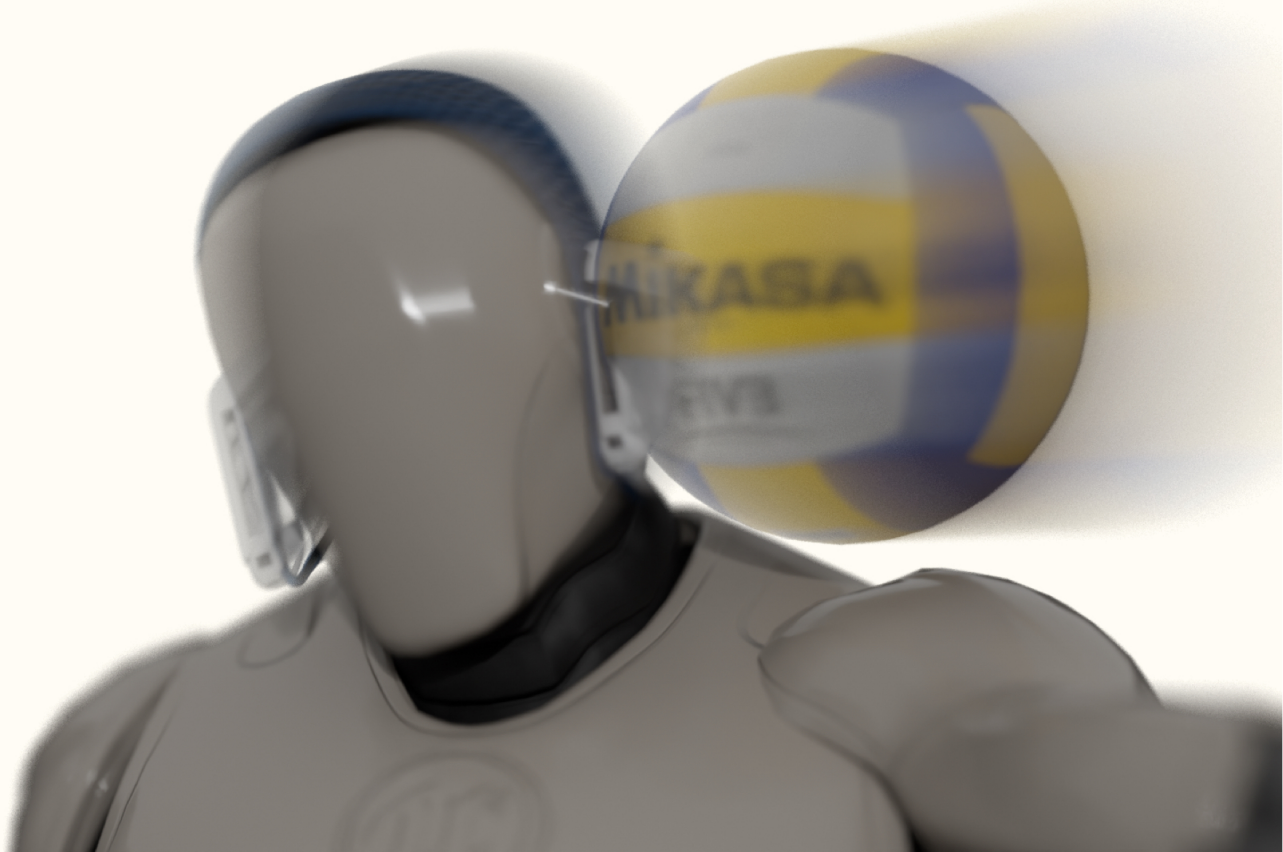


SOFTENING IMPACTS

Larger Surface Area:
A larger surface area allows for better distribution of force on the head, ensuring the impact is felt less by the user and reducing the change of the guard breaking from the hit.



Silicone Padding:
An increased density of silicone in the area ensures the blow is cushioned as much as possible. Instead of having hard plastic like in a normal cap, the mix of silicone and soft fabric ensures absorption of force.



ANALYSIS

Feedback from players:
Talking to fellow players, the feedback for the product was generally very positive:

“I like that the ear protection is flat it makes it look less lame” - Joe

"Standard caps have so many design flaws, but these small tweaks really fix a lot of them. Better fit, better comfort, and way less hassle during a match." - Hannah M.

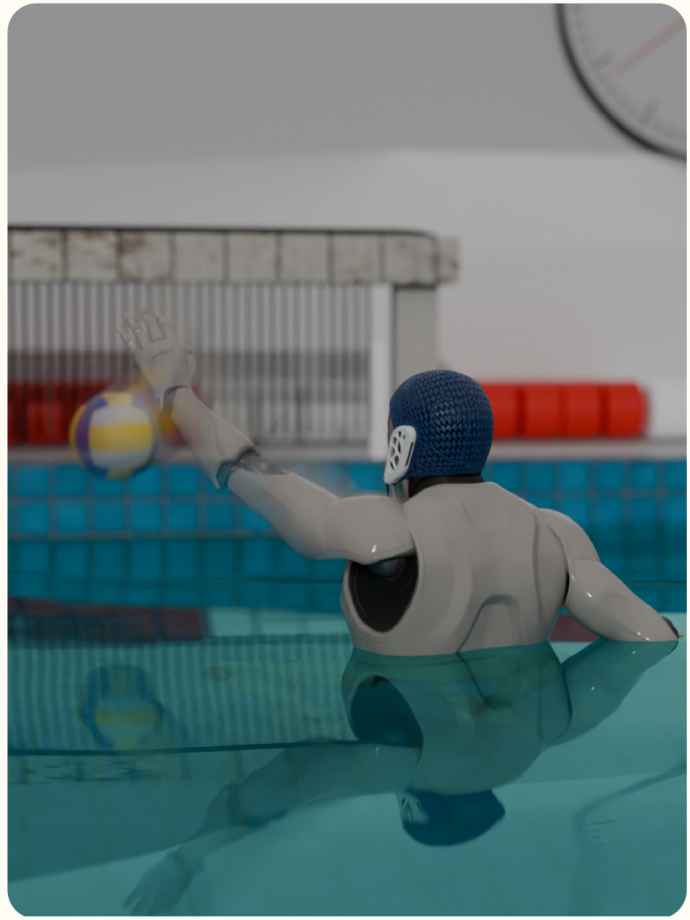
"More comfort and better sound. Small changes, big impact." - Hannah D.

Potential Improvements:
When getting feedback from them, some very apparent flaws and things to change came to us which could be improved on. These can be summarised in 3 key ideas:

Adding a silicone strip on the forehead to stop it from slipping if someone yanks on it.

Accommodating for different head-shapes or hair lengths as the cap can be quite restrictive, especially with the added silicone.

Making the ear guards replaceable if they break and so you can clean it if needed.



BACKGROUND

A rough sport

Waterpolo caps are at heart, meant to protect your ears from being burst by what can be a **very violent sport**. Between people kicking your head, balls being hurdled into it, and even punches being thrown, there are plenty of ways to get injured.



Key Pain Points

While the function of the caps is very simple, they are not so well designed that no improvements can be made. Many in the waterpolo community end up dealing with the same issues:

Injury ⟶ The ear guard often injure more than protect

Comfort → The caps can be uncomfortable to wear for long periods

Hear-ability ⟶ Often, you cannot hear the referee or teammates

Intuativity ⟶ Like a USB, it's hard to place them first time

Key features

Broken Ear guard:
Meant to protect the ear from major injuries from other player or ball impacts.

Player Number:
Informs the referee of who scored or committed a foul.



Fabric Knot:
Holds the cap on the player's head, avoiding it being ripped off.

Holes:
Allows some sound to get through and for water to flow out.

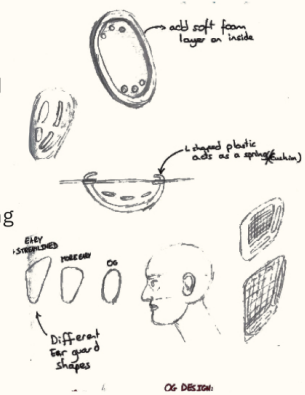
Cap Fabric:
Holds hair back stopping them from going in people's faces.

SKETCHING

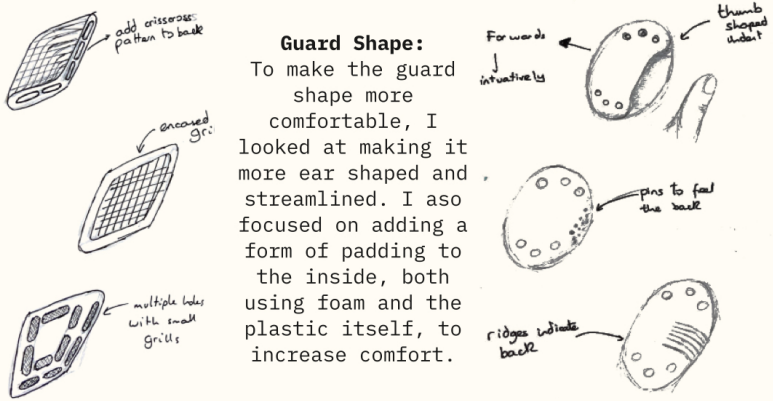
Experimentation

Early sketches can be split into **3 key parts** around the ear guard. Firstly, Around the grill to increase hear-ability, while maintaining protection; then the overall shape to increase comfort, and finally around the intuitiveness of wear.

Grill Shape:
The aim of these sketches was to add both a flair of interest to the design, while also improving the hear-ability and reducing water entrapment. Three variations around possible grill looks were tested:



Intuitiveness:
For the user to be able to place the cap correctly at first touch, experiments around the shape and how a user feels and reacts to it were done. I looked at both the overall shape and adding textures:



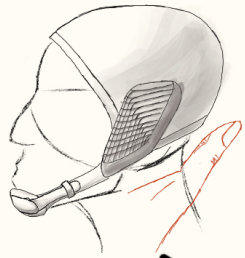
Guard Shape:
To make the guard shape more comfortable, I looked at making it more ear shaped and streamlined. I aso focused on adding a form of padding to the inside, both using foam and the plastic itself, to increase comfort.

Refined Sketches

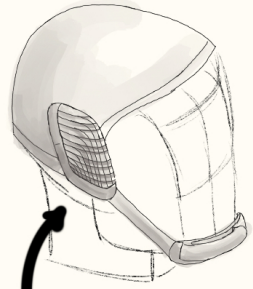
Improved clasp:
Ensures it does not come undone and easier to adjust. This had to be abandoned due to regulations.

Chin protector:
Improves protection, especially from kicks when swimming after someone.





Intuitive shape:
Fits thumb perfectly giving direction to the cap.



Improved grill:
Improved hear-ability through a better designed grill with more holes.

CAD PROTOTYPES

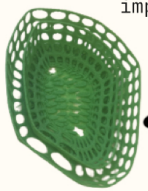
Early CAD models

CAD models were created both parametrically and generatively to experiment on how both options would look, but also to then 3**D print** them to be able to experiment with them in real life. All of these models were built in Blender, with the Voronoi structure below having been made using geometry nodes.

Moulded shape:
Moulded ear guard shape allows for increased comfort against head.



Voronoi Structure:
Increased strength allows for better impact resistance and much better noise permeability.





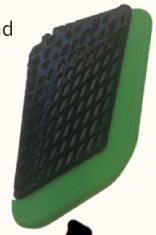
Looks:
Improved looks to a classical flat design makes it stand out more as a design.



Issues:
Issues around water getting stuck in the smaller holes due to surface tension



Wireframe Structure:
Increased strength allows for better impact resistance and much better noise permeability.



Streamlining:
Reduces the effects of water resistance and allows more space for the ear, increasing comfort.

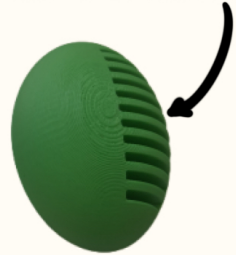
Dented:
Most comfortable to use and easiest to feel, especially when hands are wet.



Dimples:
Too difficult to feel, especially when hands are wet.



Slots:
Slots are easier to feel than dimples, but can sometimes be hard to in aninstant.



DEVELOPMENT

Generative AI

Generative AI was used to be able to connect the different ideas and designs from the earlier phase. From this, I decided I wanted to keep the Voronoi grill design, extending it to be a **cover for the fabric** as well, but incorporating it into the streamlined structure. This also allowed me to add a dent to the back for better orientation perception.

Voronoi Netting:
Adds a level of uniqueness to the design.



Voronoi Structure:
Ear guard with a thicker Voronoi structure to improve strength.

Above Fabric:
Ensures hair is held in place while improving looks.



Integration:
Integrated into the streamlined ear guard shape.

3D Modelling



Early experiments with adding a **voronoi mesh above** the fabric proved to be positive. This mesh would be made out of **silicone or rubber** to give better protection to the head and structure to the cap.

From this, I decided to experiment with making the Voronoi **attach directly to the ear guard**, in order to flow between the two pieces better giving an overall smoother look. However, I also wanted the ear guard to look **more solid** in order to counteract the unevenness of the mesh above the fabric.

